

Which State-Level Pharmacy Markets Get Generics First After Launch? An ML.NET-Based Forecasting Framework for Generic Drug Diffusion and Access Inequality

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Abstract

Generic drug approvals are expected to improve medication access, increase competition, and reduce treatment cost. However, regulatory approval does not necessarily mean that a generic medicine becomes visible across all pharmacy markets immediately after launch. Some generic products may first appear in high-volume markets, while others may spread broadly across many markets, remain delayed, or remain non-observed in public utilization data for several quarters. This study proposes a reproducible data engineering and machine learning framework to measure and forecast generic drug diffusion after first generic approval using Numeric Distribution (ND), Weighted Distribution (WD), and ML.NET.

The study uses FDA first-generic approval records as launch-start proxies and Medicaid State Drug Utilization Data as a public proxy for state-level pharmacy access markets. A SQL Server pipeline was developed to clean first-generic launch records, import Medicaid utilization records, match brand and generic product names, calculate quarterly state-level diffusion metrics, and generate ML.NET training data. Numeric Distribution measures the percentage of active state-level molecule markets where generic utilization is observed. Weighted Distribution measures the share of total molecule prescription volume represented by the states where generic utilization is observed. The difference between WD and ND, called Access Gap, is used to identify whether generics reached high-volume markets before broad geographic diffusion.

The final dataset produced 1,188 quarterly ND/WD observations and 444 first-four-quarter launch observations across 111 complete generic launch cases. The ML.NET training dataset contained 333 rows, with 213 Fast adoption rows, 12 Medium adoption rows, and 108 Slow adoption rows. The ML.NET multiclass classification model achieved 95.18% micro accuracy, 66.67% macro accuracy, and 0.2232 log loss. The lower macro accuracy reflects the limited number of Medium examples and is an important indicator of class imbalance.

The results reveal several meaningful launch diffusion patterns. Droxidopa showed the largest high-volume-first pattern, with Q1 ND of 16.67% but WD of 84.30%, producing a 67.63-point Access Gap. Tasimelteon showed a similar pattern, reaching only 14.29% ND in Q2 but 76.29% WD, meaning one observed state represented most molecule prescription volume. Dabigatran etexilate showed persistent high-volume-first diffusion, remaining below 50% ND while maintaining WD above 84% during the first four quarters. Liraglutide, acetaminophen/ibuprofen, amantadine, amlodipine, budesonide/formoterol, and several other launches showed immediate broad and high-volume diffusion. Medium adoption cases included darunavir, deflazacort, pazopanib, and saxagliptin. Several slow cases showed no observed generic diffusion in the first four quarters under the current public-data matching method.

This study demonstrates that ND and WD provide richer insight than prescription volume alone. A generic launch can appear limited by numeric state coverage but still cover most prescription volume if it reaches high-volume states first. The paper contributes a .NET-based health informatics pipeline, an ND/WD measurement framework for public pharmaceutical data, and an ML.NET model for classifying generic adoption patterns. The study should be interpreted as an applied computer science and data engineering contribution rather than a clinical or pharmacological evaluation.

Keywords: generic drug launch, numeric distribution, weighted distribution, ML.NET, SQL Server, Medicaid drug utilization, health informatics, pharmaceutical data mining, access inequality, generic diffusion, C#, applied machine learning

1. Introduction

Generic medicines are central to pharmaceutical access and affordability. When a first generic version of a branded medicine receives regulatory approval, the market expects increased competition, wider access, and possible cost reduction. However, regulatory approval alone does not fully answer an important practical question: where does the generic appear first?

A first generic medicine may not reach all pharmacy markets at the same time. It may first appear in large states with high prescription volume. It may spread immediately across almost all active state markets. It may remain delayed in Medicaid utilization records. It may also be approved but not become visible in public utilization data during the early post-approval quarters. These patterns matter because generic diffusion is not only about whether a product exists, but also about where it becomes observable and how important those markets are.

This study asks:

Which state-level pharmacy access markets show generic utilization first after launch?

Because this research uses public state-level data rather than private individual pharmacy-level sales or inventory data, the technical version of the question is:

Which state-level pharmacy access markets show generic utilization first after launch, and do those markets represent high prescription volume?

The paper adapts two pharmaceutical distribution concepts: Numeric Distribution and Weighted Distribution. Numeric Distribution measures how broadly a product appears across markets. Weighted Distribution measures how important those markets are in terms of volume. A product can have low ND but high WD if it appears in only a few markets that represent most prescription demand. This pattern is especially interesting in generic launch analysis because it can indicate high-volume-first diffusion.

This research is designed from a computer science and health informatics perspective. The focus is not to evaluate clinical effectiveness, drug safety, or medical appropriateness. Instead, the contribution is a reproducible data pipeline and machine learning framework using SQL Server, C#, and ML.NET to study generic launch diffusion from public data.

The study contributes:

1. A public-data method for measuring generic launch diffusion using ND and WD.
2. A SQL Server pipeline for transforming FDA and Medicaid data into quarterly diffusion metrics.

3. A classification framework that predicts Fast, Medium, or Slow generic adoption using early post-launch signals.
4. An ML.NET implementation suitable for production-oriented .NET healthcare analytics environments.
5. A set of empirical observations showing high-volume-first, broad/high-volume, medium, slow, and non-observed diffusion patterns.

2. Background

2.1 First generic drug approvals

First generic approvals are important because they represent the first FDA approval permitting a manufacturer to market a generic drug product in the United States [1]. These approvals can introduce competition into markets previously served by branded products. However, approval date is not necessarily the same as real-world commercial availability date. Therefore, this study treats the FDA approval date as a **launch-start proxy**, not as proof that the product was available in pharmacies on that exact date.

This distinction is important for interpreting early zero-diffusion cases. If a generic is not visible in Medicaid data during the first one or two quarters after approval, it does not automatically mean that the launch failed. It may reflect delayed commercial availability, naming mismatch, Medicaid capture limitations, data suppression, or market-specific reimbursement behavior.

2.2 Medicaid State Drug Utilization Data

Medicaid State Drug Utilization Data provides state-level outpatient drug utilization information, including drug name, National Drug Code, number of prescriptions, units reimbursed, and reimbursement amounts [6], [7]. In this study, Medicaid prescription count is used as the main volume measure.

The analysis uses state-level markets as a proxy for pharmacy access markets. This means that a “market” in the study is not an individual pharmacy store. It is a state-level Medicaid utilization market for a specific brand/generic molecule.

2.3 Why ND and WD matter

A simple count of generic prescriptions can show whether utilization exists, but it does not explain the structure of diffusion. For example, two generics may have similar early prescription totals but very different diffusion patterns:

- Generic A may appear in many states but mostly low-volume states.
- Generic B may appear in only a few states but those states may represent most molecule volume.

Numeric Distribution and Weighted Distribution can separate these patterns.

Numeric Distribution answers:

How many active markets have the generic?

Weighted Distribution answers:

How important are the markets where the generic appears?

The difference between them gives the Access Gap:

$$\text{Access Gap} = \text{WD} - \text{ND}$$

A high positive Access Gap suggests that the generic reached high-volume markets before achieving broad geographic coverage.

3. Research Questions

The main research question is:

Can state-level Numeric Distribution (ND) and Weighted Distribution (WD) metrics identify whether first-generic drugs spread evenly across pharmacy access markets after approval or initially concentrate in high-volume markets?

The study also asks:

1. Can ND and WD identify diffusion patterns that are not visible from prescription volume alone?
2. Which generic launches show high-volume-first diffusion?
3. Which launches show broad and high-volume diffusion?
4. Which launches remain slow or non-observed in early Medicaid utilization data?
5. Can early ND/WD features from quarters 1–3 classify Q4 adoption as Fast, Medium, or Slow?
6. Is ML.NET suitable for building a production-oriented pharmaceutical analytics pipeline in the .NET ecosystem?

4. Data and Study Scope

4.1 Data sources

The study uses two public data sources:

1. FDA first-generic approval records [1]–[5].
2. Medicaid State Drug Utilization Data [6], [7].

The FDA data provides the launch candidates, brand/generic product names, approval dates, and manually assigned therapeutic grouping. Medicaid data provides quarterly state-level utilization records.

4.2 Study period

The current study uses launches and Medicaid utilization data available in the project pipeline from 2021 onward. The first-generic launch candidates were selected primarily from 2021–2024 FDA first-generic approval lists so that the post-launch quarters could be observed in the available Medicaid utilization data.

4.3 Unit of analysis

The unit of analysis is the **state-level molecule market**.

A molecule market is defined using matched brand and generic product names. For each launch, the pipeline identifies utilization rows related to the brand name and generic ingredient. A state is considered

active if brand or generic molecule utilization is present. A state is considered generic-observed if generic utilization is detected.

4.4 Scope limitations

The study does not use private pharmacy sales, inventory, or store-level data. Therefore, it cannot directly say whether CVS, Walgreens, Walmart, independent pharmacies, specialty pharmacies, or mail-order pharmacies received generics first. It can only identify state-level Medicaid utilization patterns.

5. Methodology

Technical scope. The implementation was developed using SQL Server for data staging, cleaning, matching, and quarterly metric calculation, and C#/ML.NET for multiclass adoption prediction. The study is positioned as an applied computer science and health informatics project rather than a clinical or pharmacological evaluation.

5.1 Overall pipeline

The project uses an end-to-end SQL Server and ML.NET workflow.



Figure 1. End-to-end system architecture for the generic launch diffusion pipeline, showing public FDA and Medicaid inputs, SQL Server staging, data cleaning, brand/generic matching, quarterly ND/WD metric calculation, ML.NET training, and final research outputs.

The pipeline includes:

1. Import FDA first-generic launch candidates into a staging table.
2. Import Medicaid SDUD CSV files into yearly staging tables.
3. Load cleaned launch candidates into FirstGenericLaunches.
4. Load cleaned Medicaid records into MedicaidUtilization.
5. Generate automatic brand/generic match terms.
6. Match Medicaid product names to launch candidates.
7. Calculate quarterly ND, WD, and Access Gap.

8. Create ModelTrainingData.
9. Export model_training.csv.
10. Train an ML.NET multiclass classification model.
11. Analyze results in Excel tables and figures.

5.2 Launch-start proxy

For each generic launch, the FDA approval date is converted to an approval year and approval quarter. The approval quarter is treated as QuarterSinceLaunch = 1.

The formula is:

$$\text{QuarterSinceLaunch} = ((\text{UtilizationYear} - \text{ApprovalYear}) \times 4) + (\text{UtilizationQuarter} - \text{ApprovalQuarter}) + 1$$

Example:

If a generic was approved in 2023 Q2:

- 2023 Q2 = Q1 after launch
- 2023 Q3 = Q2 after launch
- 2023 Q4 = Q3 after launch
- 2024 Q1 = Q4 after launch

5.3 Numeric Distribution

Numeric Distribution measures geographic breadth of observed generic utilization.

Let:

- ActiveStates = number of states with brand or generic molecule utilization
- GenericStates = number of active states where generic utilization is observed

Then:

$$\text{ND} = (\text{GenericStates} / \text{ActiveStates}) \times 100$$

Example:

If 10 active states have molecule utilization and 2 states have generic utilization:

$$\text{ND} = 20\%$$

5.4 Weighted Distribution

Weighted Distribution measures volume-weighted importance of the states where generic utilization appears.

Let:

- MoleculePrescriptions = total brand/generic molecule prescriptions across all active states

- $\text{MoleculePrescriptionsInGenericStates}$ = total molecule prescriptions in states where generic utilization is observed

Then:

$$\text{WD} = (\text{MoleculePrescriptionsInGenericStates} / \text{MoleculePrescriptions}) \times 100$$

Example:

If a generic appears in only 2 out of 10 states, but those 2 states represent 75% of molecule prescriptions:

$$\text{ND} = 20\%$$

$$\text{WD} = 75\%$$

This indicates high-volume-first diffusion.

5.5 Access Gap

Access Gap measures the difference between weighted and numeric coverage.

$$\text{Access Gap} = \text{WD} - \text{ND}$$

Interpretation:

- **Large positive Access Gap:** generic reached high-volume states before broad geographic diffusion.
- **High ND and high WD:** generic diffused broadly and reached high-volume states.
- **Low ND and low WD:** weak or non-observed early diffusion.
- **Negative Access Gap:** generic reached many states but missed high-volume markets. This was not a dominant pattern in the current results.

5.6 Adoption class

Each complete launch was assigned an adoption class based on Q4 WD:

- **Fast:** $\text{Q4 WD} \geq 80\%$
- **Medium:** $50\% \leq \text{Q4 WD} < 80\%$
- **Slow:** $\text{Q4 WD} < 50\%$

This classification is volume-weighted. A launch can be considered Fast even if ND is below 80%, as long as the generic reaches states that represent at least 80% of total molecule prescription volume.

5.7 ML.NET model

The ML task is multiclass classification. The model predicts adoption class using early launch-quarter features.

Input features:

- QuarterSinceLaunch
- CurrentND

- CurrentWD
- PreviousND
- PreviousWD
- NDGrowth
- WDGrowth
- AccessGap
- TotalGenericPrescriptions
- TotalClassPrescriptions / molecule prescriptions

Output label:

- Fast
- Medium
- Slow

The ML.NET pipeline maps the label to a key, concatenates numeric features, normalizes features, trains a multiclass classifier, and maps predicted labels back to text [8]–[12].

6. Results

6.1 Dataset summary

The expanded dataset is substantially stronger than the first prototype.

Table 1. Dataset summary of the generic launch diffusion study.

Metric	Value
Total quarterly ND/WD observations	1,188
First-four-quarter observations	444
Complete Q1–Q4 launch cases	111
ML.NET training rows	333
Fast training rows	213
Medium training rows	12
Slow training rows	108
Adoption classes present	Fast, Medium, Slow

At the launch level, the 111 complete cases produced:

Table 2. Launch-level adoption class distribution based on fourth-quarter Weighted Distribution.

Adoption Class	Launch Count	Share
Fast	71	64.0%
Medium	4	3.6%
Slow	36	32.4%

At the ML training-row level, each complete launch contributes three early quarters, so the training rows were:

Table 3. ML.NET training-row distribution by adoption class.

Adoption Class	Training Rows	Share
Fast	213	64.0%
Medium	12	3.6%
Slow	108	32.4%

The expanded dataset provides a substantially larger basis for exploratory applied analysis than the initial prototype. However, the Medium class remains underrepresented.

6.2 Overall first-four-quarter diffusion behavior

Across the 444 first-four-quarter observations, the average ND and WD increased over time.

Table 4. Overall quarterly ND/WD summary during the first four post-approval quarters.

Quarter Since Launch	Mean ND	Median ND	Mean WD	Median WD	Mean Access Gap
Q1	49.21%	36.00%	57.65%	82.58%	8.44
Q2	56.78%	79.17%	63.68%	95.19%	6.91
Q3	59.69%	87.50%	65.35%	97.64%	5.66
Q4	60.41%	90.91%	65.67%	98.84%	5.27

The median WD values are much higher than the mean WD values because the dataset contains two major groups: launches with rapid/broad observed diffusion and launches with zero or weak observed diffusion. The high median WD suggests that among many launches where generic diffusion is observed, the generic tends to reach important prescription-volume states quickly.

When considering only observations where generic utilization was detected, the diffusion pattern becomes even clearer.

Table 5. Detected-only quarterly ND/WD summary during the first four post-approval quarters.

Quarter Since Launch	Mean ND	Median ND	Mean WD	Median WD	Mean Access Gap
Q1	79.17%	100.00%	92.75%	100.00%	13.58
Q2	85.17%	100.00%	95.53%	100.00%	10.36
Q3	88.33%	100.00%	96.71%	100.00%	8.38
Q4	89.40%	100.00%	97.19%	100.00%	7.79

This detected-only summary shows that once a generic appears in Medicaid utilization data, it often reaches high-volume states quickly.

6.3 Diffusion interpretation categories

Using the interpretation rules from the SQL output, the 444 first-four-quarter observations were classified as:

Table 6. Diffusion interpretation categories across first-four-quarter observations.

Interpretation	Count
Broad and high-volume diffusion	237
Weak or non-observed early generic diffusion	151
Mixed diffusion pattern	35
High-volume markets got generic first	21

This distribution shows that broad/high-volume diffusion was the most common observed pattern. However, the 21 high-volume-first cases are especially important because they demonstrate why WD is necessary. These cases would be underestimated if only Numeric Distribution were used.

7. Key Empirical Findings

7.1 Finding 1: High-volume-first diffusion is clearly visible

The strongest observation is that several generics reached high-volume states before they achieved broad geographic coverage.

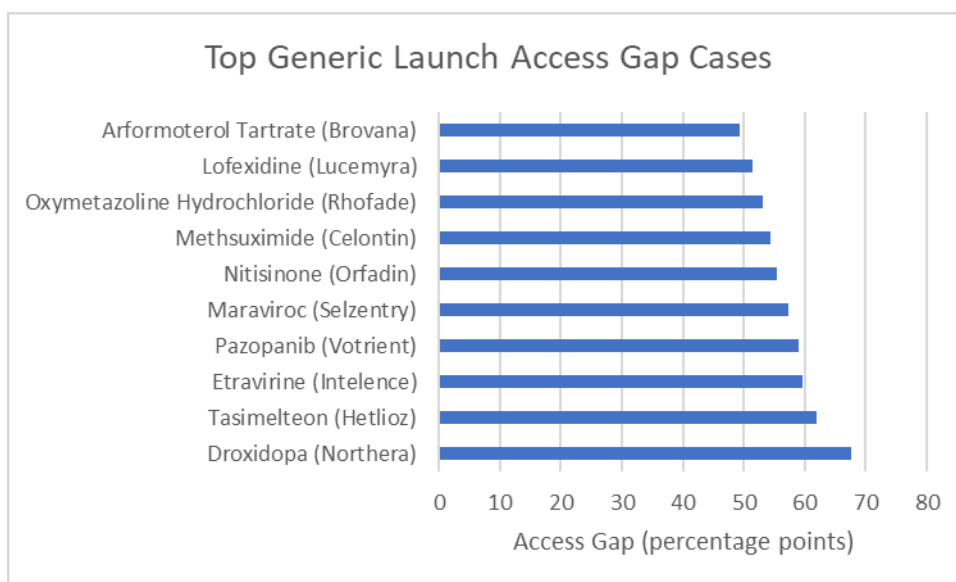


Figure 2. Top high-volume-first generic launch cases ranked by each drug's maximum Access Gap during the first four post-approval quarters.

The top Access Gap cases included:

Table 7. Top high-volume-first generic launch cases ranked by maximum Access Gap during the first four post-approval quarters.

Generic	Quarter Since Launch	ND (%)	WD (%)	Access Gap
Droxidopa (Northera)	Q1	16.67	84.30	67.63
Tasimelteon (Hetlio)	Q2	14.29	76.29	62.00
Etravirine (Intelence)	Q1	7.69	67.26	59.57
Pazopanib (Votrient)	Q2	20.00	78.88	58.88
Maraviroc (Selzentry)	Q1	17.65	74.99	57.34
Nitisinone (Orfadin)	Q3	33.33	88.78	55.45
Methsuximide (Celontin)	Q1	16.67	71.08	54.42
Oxymetazoline Hydrochloride (Rhofade)	Q3	31.25	84.29	53.04
Lofexidine (Lucemyra)	Q1	22.22	73.56	51.34
Arformoterol Tartrate (Brovana)	Q1	10.64	60.05	49.41

These examples show why the title question matters. Generics may not reach most state-level pharmacy access markets immediately, but they may reach the states that represent most prescription volume.

7.2 Finding 2: Droxidopa had the largest observed Access Gap

Droxidopa showed the largest high-volume-first signal in the dataset.

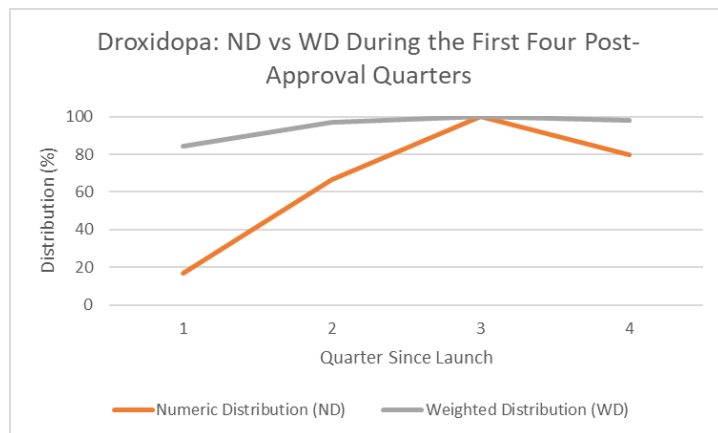


Figure 3. Droxidopa ND and WD trend during the first four post-approval quarters, showing the largest observed high-volume-first diffusion pattern in the dataset.

In Q1 after launch:

- ND = 16.67%
- WD = 84.30%
- Access Gap = 67.63

This means generic utilization was observed in only 1 out of 6 active state markets, but that state represented 84.30% of molecule prescription volume. This is a strong high-volume-first pattern.

By Q4, droxidopa reached 97.90% WD and was classified as Fast adoption. The pattern suggests that early limited numeric coverage did not necessarily mean low market importance. A small number of state markets captured most molecule volume.

7.3 Finding 3: Tasimelteon showed one of the clearest high-volume-first stories

Tasimelteon is one of the most interesting case studies because its early diffusion pattern is easy to explain.

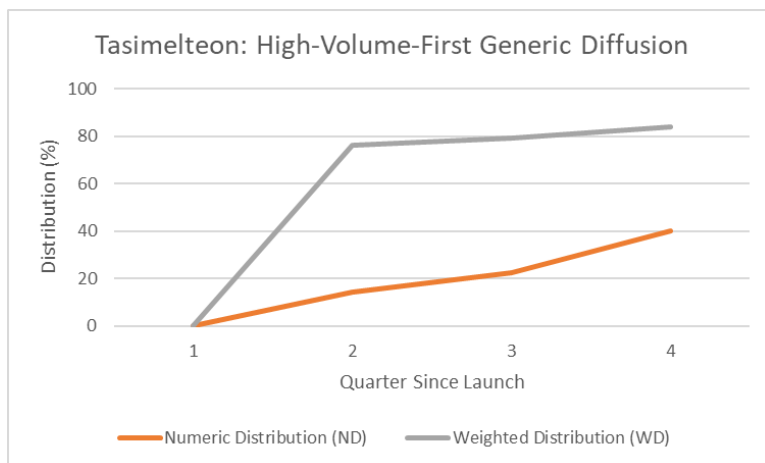


Figure 4. Tasimelton ND and WD trend during the first four post-approval quarters, showing low numeric state coverage but high weighted prescription-volume coverage after generic utilization became observable.

Table 8. Tasimelton quarterly ND/WD diffusion metrics.

Quarter Since Launch	Active States	Generic States	ND	WD	Access Gap
Q1	11	0	0.00%	0.00%	0.00
Q2	7	1	14.29%	76.29%	62.00
Q3	9	2	22.22%	78.96%	56.73
Q4	10	4	40.00%	84.07%	44.07

Tasimelton had no observed generic diffusion in Q1. In Q2, only one active state had generic utilization, but that state represented 76.29% of molecule prescriptions. By Q4, generic utilization was still visible in only 40.00% of active states, but those states represented 84.07% of molecule volume.

This is a classic high-volume-first diffusion pattern. If the study used only ND, the launch might appear weak. WD reveals that the generic reached highly important markets early.

7.4 Finding 4: Dabigatran etexilate had persistent high-volume-first diffusion

Dabigatran etexilate showed a stable high-volume-first pattern across all four quarters.

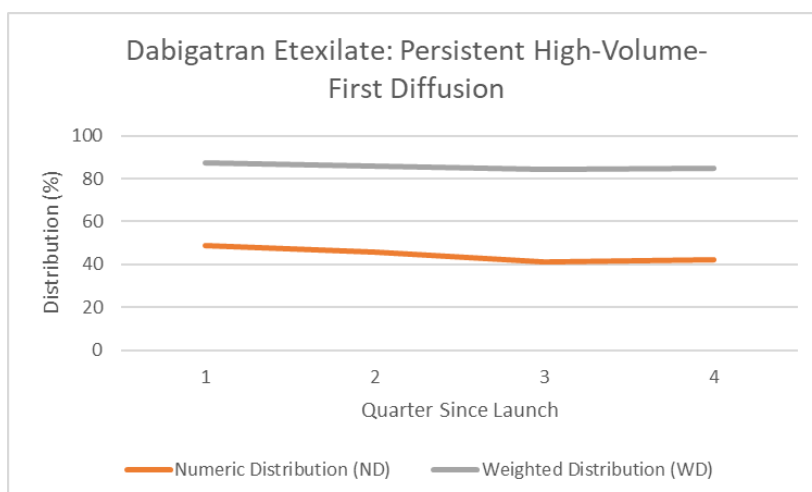


Figure 5. Dabigatran etexilate ND and WD trend during the first four post-approval quarters, showing persistent high-volume-first diffusion with WD remaining above ND across all observed quarters.

Table 9. Dabigatran etexilate quarterly ND/WD diffusion metrics.

Quarter Since Launch	Active States	Generic States	ND	WD	Access Gap
Q1	47	23	48.94%	87.57%	38.63
Q2	46	21	45.65%	85.96%	40.31
Q3	46	19	41.30%	84.24%	42.94
Q4	45	19	42.22%	84.99%	42.76

This case is important because generic dabigatran did not reach most active states. ND stayed around 41–49%. However, WD stayed around 84–88%. This means that less than half of state markets represented most prescription volume.

Dabigatran is one of the best examples for explaining why a weighted distribution metric is necessary in generic diffusion research.

7.5 Finding 5: Some launches diffused broadly and immediately

A different group of launches showed immediate broad and high-volume diffusion. These launches had ND and WD near 100% from Q1.

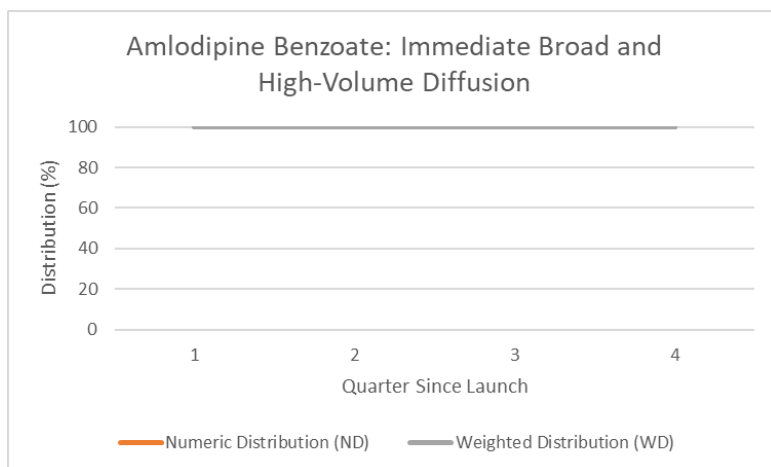


Figure 6. Broad and high-volume diffusion example for amlodipine benzoate, showing generic utilization across many active state-level markets while also covering high prescription-volume states during the first four post-approval quarters.

Examples included:

- Acetaminophen and ibuprofen
- Amantadine
- Amlodipine benzoate
- Aripiprazole
- Azelastine hydrochloride
- Baclofen
- Brimonidine
- Budesonide
- Budesonide and formoterol fumarate
- Carbidopa, levodopa and entacapone
- Ciprofloxacin
- Diclofenac
- Digoxin
- Doxycycline
- Drospirenone
- Enalapril maleate
- Fenoprofen calcium
- Halobetasol propionate

- Ibuprofen and famotidine
- Levothyroxine sodium
- Liraglutide

The dataset contained 47 launches with Q1 ND and Q1 WD both at or above 90%. This suggests that for many products, generic utilization was visible broadly and in high-volume states immediately after the approval quarter.

This pattern may reflect products where Medicaid already recorded utilization under ingredient-based product naming, rapid product substitution, broad pharmacy availability, or product-name matching characteristics. Because the method uses public product names, these cases must be interpreted as observed state-level utilization diffusion rather than proof of specific store-level stocking.

7.6 Finding 6: Liraglutide showed near-complete diffusion

Liraglutide is a useful broad-diffusion case.

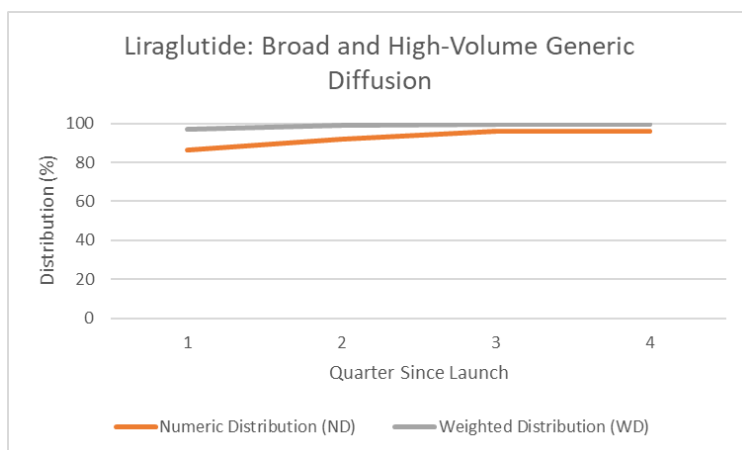


Figure 7. Liraglutide ND and WD trend during the first four post-approval quarters, showing rapid broad and high-volume generic diffusion.

Earlier prototype results showed liraglutide quickly reaching high state coverage and high weighted coverage. In the expanded dataset, it remains a strong example of broad and high-volume diffusion. This is useful as a contrast to tasimelteon and dabigatran. Tasimelteon and dabigatran show high-volume-first diffusion with lower ND, while liraglutide shows broad/high-volume diffusion.

7.7 Finding 7: Medium adoption now exists but is rare

The expanded dataset fixed the earlier weakness where Medium adoption was missing. However, Medium is still rare.

Table 10. Medium adoption cases based on fourth-quarter Weighted Distribution.

Generic	Brand	Therapeutic Class	Q1 WD	Q2 WD	Q3 WD	Q4 WD
Darunavir	Prezista	HIV / Antiviral	0.00	0.00	0.00	62.33
Deflazacort	Emflaza	CNS / Neuromuscular	79.68	80.21	83.68	77.69
Pazopanib	Votrient	Oncology	76.04	78.88	80.41	79.47
Saxagliptin	Onglyza	Diabetes	57.02	61.30	71.54	75.34

These four launches are important because they sit between clear Fast and clear Slow patterns. Saxagliptin is especially interesting because WD increased gradually from 57.02% to 75.34%, showing a progressive medium diffusion pattern. Darunavir is different: it had no observed WD in Q1–Q3 but reached 62.33% in Q4, suggesting delayed observation rather than steady growth. Deflazacort and pazopanib remained near the Fast threshold but ended below 80% in Q4.

The small Medium class explains why macro accuracy is lower than micro accuracy in the ML.NET model.

7.8 Finding 8: Slow cases often had zero observed generic diffusion

There were 36 Slow launches at the launch level. Most of these showed Q4 WD of 0.00%.

Examples included:

- Apremilast / Otezla
- Azilsartan medoxomil / Edarbi
- Baricitinib / Olumiant
- Brexpiprazole / Rexulti
- Brivaracetam / Briviact
- Cariprazine / Vraylar
- Dapagliflozin / Farxiga
- Dronedarone / Multaq
- Empagliflozin / Jardiance
- Ertugliflozin / Steglatro
- Fidaxomicin / Dificid
- Fingolimod / Gilenya
- Ivabradine / Corlanor

- Linaclotide / Linzess
- Linagliptin / Tradjenta
- Mirabegron / Myrbetriq
- Sacubitril and valsartan / Entresto
- Valbenazine / Ingrezza
- Vortioxetine / Trintellix

These should not be described as failed launches. The safer term is:

slow, delayed-observed, or non-observed generic diffusion under the current Medicaid product-name matching method.

Possible explanations include:

1. The generic was approved but not commercially available immediately.
2. Medicaid records did not capture the generic within the first four quarters.
3. Product naming differed from the generated match term.
4. Claims were suppressed.
5. Brand utilization remained dominant.
6. Specialty or restricted distribution channels limited Medicaid capture.
7. The product was listed under NDC or manufacturer naming not captured by the current matching rule.
8. Some drugs may have ingredient-level or product-level complexity requiring RxNorm, Orange Book, or NDC validation.

7.9 Finding 9: Nitisinone is an anomalous case

Nitisinone is especially interesting because it had high WD during Q1–Q3 but dropped to Q4 WD of 0.00%, causing it to be labeled Slow.

Table 11. Nitisinone anomalous diffusion pattern.

Generic	Brand	Q1 WD	Q2 WD	Q3 WD	Q4 WD	Class
Nitisinone	Orfadin	92.75	86.44	88.78	0.00	Slow

This pattern may reflect data suppression, discontinuity in product-name matching, very small active-state denominators, rare disease market behavior, or a true utilization drop. It should be manually reviewed before final publication. This is a strong example of why automated matching must be supplemented with manual validation.

8. ML.NET Model Results

8.1 Training dataset

The ML.NET training-row distribution is shown in Table 3. The Medium class is present but underrepresented, which affects macro-level model performance.

8.2 Model performance

The ML.NET console output showed:

Table 12. ML.NET multiclass classification performance.

Metric	Value
Training rows	333
Classes detected	Fast, Medium, Slow
Micro Accuracy	95.18%
Macro Accuracy	66.67%
Log Loss	0.2232
Example prediction	Fast

The micro accuracy is high, but macro accuracy is much lower. This is expected because Medium is underrepresented. Micro accuracy is dominated by the larger Fast and Slow classes, while macro accuracy gives equal weight to each class. Therefore, the lower macro accuracy is not a weakness to hide. It is actually an important honest signal that the Medium class needs more examples or improved validation.

8.3 Interpretation of model performance

The model should be interpreted as a successful prototype classifier, not as final proof of generalizable prediction. The current features are strongly related to the Q4 WD label because the label itself is derived from weighted distribution. Therefore, the model may learn diffusion separation effectively, but stronger validation is still needed.

For a publication-quality version, the next improvements should include:

1. Cross-validation.
2. Train/test split by approval year.
3. Confusion matrix.
4. Per-class precision, recall, and F1-score.
5. Sensitivity analysis using alternative thresholds.
6. Manual validation of selected high-impact product matches.

7. NDC- or RxNorm-supported matching.

9. Discussion

9.1 ND and WD reveal patterns that raw volume cannot

The main value of the study is not simply the ML model. The main value is the ND/WD measurement framework. Prescription volume alone cannot distinguish whether a generic is broadly distributed or concentrated in a few important states. ND and WD together reveal that difference.

For example, tasimelteon in Q2 had only 14.29% ND but 76.29% WD. That means one state-level market represented most molecule prescription volume. A simple state count would make the launch look weak, while WD shows it reached a high-volume market.

9.2 High-volume-first diffusion is a real and important category

Droxidopa, tasimelteon, etravirine, pazopanib, maraviroc, methsuximide, oxymetazoline, lacosamide, and dabigatran etexilate all showed meaningful Access Gaps. These cases suggest that the first visible generic markets are often not random. They may be high-volume markets.

This has practical meaning. If a generic reaches high-volume states first, then many patients may benefit even before the product appears broadly across all active states. Conversely, if a generic spreads broadly but misses high-volume markets, the access impact may be smaller. In the current dataset, the latter pattern was not dominant.

9.3 Broad/high-volume diffusion is common

The largest interpretation category was broad and high-volume diffusion, with 237 observations. Many launches reached both high ND and high WD quickly. This may indicate rapid generic visibility in Medicaid data, broad product substitution, common retail use, or ingredient-level product naming that makes generic utilization visible across many states.

This is an encouraging pattern, but it also requires methodological caution. When ND and WD are both 100% from Q1, researchers should verify whether the generic match term is capturing the exact first generic product or broader same-ingredient utilization.

9.4 Slow diffusion should be interpreted carefully

The Slow class contains many high-profile drugs, including cardiovascular, diabetes, CNS, urology, oncology, and immunology products. It would be incorrect to conclude that these generic launches failed. Instead, the correct interpretation is that the current public-data method did not observe generic diffusion within the first four quarters.

For publication, the paper should use careful language:

“non-observed generic diffusion in state-level Medicaid utilization data”

rather than:

“generic unavailable” or “failed launch.”

9.5 The Medium class is small but analytically valuable

Medium cases are important because they prevent the model from becoming a simple Fast vs Slow classifier. Darunavir, deflazacort, pazopanib, and saxagliptin provide early evidence of middle diffusion behavior. However, only four Medium launches are not enough for strong per-class generalization. This explains the 66.67% macro accuracy.

Future work should add more years, improve matching, or test alternative thresholds to capture more intermediate diffusion patterns.

9.6 Why ML.NET is appropriate

This paper is written from a computer science perspective. The goal is not only exploratory analysis but also an implementable analytics pipeline. ML.NET is appropriate because it supports model training and inference inside the .NET ecosystem. For a .NET developer working in healthcare data systems, a C# and SQL Server pipeline is practical and realistic.

The study demonstrates how pharmaceutical analytics can be operationalized in an enterprise-style software architecture:

- SQL Server for ETL and metric calculation
- C# for application logic
- ML.NET for classification
- Excel/Power BI/Blazor potential for reporting
- GitHub for reproducibility

This makes the framework relevant for applied computer science, health informatics, and production-oriented healthcare analytics.

10. Limitations

This study has several important limitations.

First, the study uses state-level Medicaid utilization data, not pharmacy-store-level sales or inventory data. Therefore, it cannot identify specific pharmacy chains or individual pharmacies.

Second, FDA approval date is used as a launch-start proxy. Actual commercial launch date may differ.

Third, product matching currently relies on brand/generic name matching. This can over-match or under-match products. Some results with immediate 100% ND and WD may require validation to confirm exact first-generic relevance.

Fourth, Medicaid utilization data does not represent all payer channels. It excludes commercial insurance, Medicare-only patterns, cash-pay behavior, hospital-only usage, and some specialty distribution dynamics.

Fifth, suppressed rows are excluded, which may understate early utilization for low-volume products.

Sixth, some drugs are specialty, rare disease, injectable, ophthalmic, dermatology, or clinic-administered products. These may not behave like ordinary retail oral medications.

Seventh, the Medium class remains underrepresented. The model achieved high micro accuracy, but macro accuracy was lower because of class imbalance.

Eighth, the current model uses a random train/test split rather than time-based validation. A stronger version should train on older launches and test on newer launches.

Ninth, the study does not measure clinical outcomes, patient-level access, adherence, price reduction, or patient cost savings.

Tenth, some drug records may require RxNorm, Orange Book, or NDC-level validation before peer-reviewed publication.

11. Future Work

Future work should improve the study in several directions.

1. Add RxNorm or NDC-level normalization.
2. Validate brand/generic mapping manually for the top 20 high-impact cases.
3. Add FDA Orange Book identifiers and applicant/manufacture information.
4. Distinguish exact first generic products from broader ingredient-level utilization.
5. Add train/test split by approval year.
6. Add cross-validation.
7. Add confusion matrix and per-class precision/recall/F1-score.
8. Add regression models to predict Q4 WD directly.
9. Add time-to-threshold analysis, such as time to 50% WD and time to 80% WD.
10. Compare Medicaid results with Medicare Part D utilization.
11. Create a Blazor or ASP.NET dashboard for interactive drug-level diffusion curves.
12. Add pharmacy density or rural/urban access data if public pharmacy location datasets are used.
13. Repeat analysis separately by product type: oral retail, topical, respiratory, injectable, specialty, and rare disease.
14. Conduct sensitivity analysis using different Fast/Medium/Slow thresholds.

12. Conclusion

This study developed a reproducible .NET-based framework for measuring and forecasting generic drug diffusion after first generic approval. Using FDA first-generic approval records and Medicaid State Drug Utilization Data, the study calculated state-level Numeric Distribution, Weighted Distribution, and Access Gap for 111 complete generic launch cases.

The results show that generic diffusion is not uniform. Some generics reach high-volume markets before broad geographic coverage. Droxidopa had the largest Access Gap, with Q1 ND of 16.67% and WD of 84.30%. Tasimelteon showed a strong high-volume-first pattern, reaching only 14.29% ND but 76.29% WD in Q2. Dabigatran etexilate showed persistent high-volume-first diffusion across the first four

quarters. Other products showed immediate broad and high-volume diffusion. A third group showed slow or non-observed diffusion in Medicaid utilization data.

The ML.NET model classified generic adoption as Fast, Medium, or Slow using early ND/WD signals. The model achieved 95.18% micro accuracy and 66.67% macro accuracy on 333 training rows. The high micro accuracy suggests that early ND/WD features are useful, while the lower macro accuracy highlights Medium-class imbalance and the need for more validation.

The main contribution is not a clinical claim about medicine effectiveness. The contribution is a computer science and health informatics framework that transforms public pharmaceutical data into interpretable diffusion metrics and a production-oriented ML.NET classification pipeline. The study demonstrates that ND and WD can help answer a practical market-access question: generics may not reach every pharmacy market first, but they may reach the most important high-volume markets first.

Conflict of Interest / Disclosure

The author declares no conflict of interest. This research was conducted independently using publicly available FDA and Medicaid datasets. The analysis, interpretations, and conclusions are the author's own.

Data and Code Availability

The SQL scripts, ML.NET source code, reproducibility notes, and sample output tables are available in the project GitHub repository: <https://github.com/Tanvir8212/generic-launch-diffusion-mlnet>.

References

- [1] U.S. Food and Drug Administration. First Generic Drug Approvals. Available at: <https://www.fda.gov/drugs/drug-and-biologic-approval-and-ind-activity-reports/first-generic-drug-approvals>. Accessed: May 18, 2026.
- [2] U.S. Food and Drug Administration. 2021 First Generic Drug Approvals. Available at: <https://www.fda.gov/drugs/drug-and-biologic-approval-and-ind-activity-reports/2021-first-generic-drug-approvals>. Accessed: May 18, 2026.
- [3] U.S. Food and Drug Administration. 2022 First Generic Drug Approvals. Available at: <https://www.fda.gov/drugs/drug-and-biologic-approval-and-ind-activity-reports/2022-first-generic-drug-approvals>. Accessed: May 18, 2026.
- [4] U.S. Food and Drug Administration. 2023 First Generic Drug Approvals. Available at: <https://www.fda.gov/drugs/drug-and-biologic-approval-and-ind-activity-reports/2023-first-generic-drug-approvals>. Accessed: May 18, 2026.
- [5] U.S. Food and Drug Administration. 2024 First Generic Drug Approvals. Available at: <https://www.fda.gov/drugs/drug-and-biologic-approval-and-ind-activity-reports/2024-first-generic-drug-approvals>. Accessed: May 18, 2026.
- [6] Centers for Medicare & Medicaid Services. State Drug Utilization Data. Available at: <https://www.medicare.gov/medicaid/prescription-drugs/state-drug-utilization-data>. Accessed: May 18, 2026.

[7] Centers for Medicare & Medicaid Services. Medicaid State Drug Utilization Data Field Descriptions. Available at: <https://www.cms.gov/files/document/medicaid-state-drug-utilization-data-field-descriptions.pdf>. Accessed: May 18, 2026.

[8] Microsoft. Categorize support issues using multiclass classification with ML.NET. Available at: <https://learn.microsoft.com/en-us/dotnet/machine-learning/tutorials/github-issue-classification>. Accessed: May 18, 2026.

[9] Microsoft. SdcaMaximumEntropyMulticlassTrainer Class. Available at: <https://learn.microsoft.com/en-us/dotnet/api/microsoft.ml.trainers.sdcamaximumentropy-multiclass-trainer>. Accessed: May 18, 2026.

[10] Microsoft. Machine learning tasks in ML.NET. Available at: <https://learn.microsoft.com/en-us/dotnet/machine-learning/resources/tasks>. Accessed: May 18, 2026.

[11] Ahmed, Z., Amizadeh, S., Bilenko, M., Carr, R., Chin, W.-S., Dekel, Y., Dupre, X., Eksarevskiy, V., Erhardt, E., Eseanu, C., et al. Machine Learning at Microsoft with ML.NET. arXiv:1905.05715, 2019. Available at: <https://arxiv.org/abs/1905.05715>. Accessed: May 18, 2026.

[12] Ahmed, Z., Amizadeh, S., Bilenko, M., Carr, R., Chin, W.-S., Dekel, Y., Dupre, X., Eksarevskiy, V., Erhardt, E., Eseanu, C., et al. Machine Learning at Microsoft with ML.NET. Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2019. Available at: <https://doi.org/10.1145/3292500.3330667>. Accessed: May 18, 2026.